Year 1 AS Pure Maths

Chapter 5

Straight line graphs

Practice Booklet

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5.1 $y = mx + c$

Practice Book, printed page 37–38.

1. Work out the gradients of the line joining these pairs of points:

a. $(4, 3)$, $(8, 6)$

b. $(5, 2)$, $(7, -1)$

c. $(3p, -4p)$, $(8p, -2p)$

Hint

The gradient, m, of a line joining the points with coordinates (x_1, y_1) and (x_2, y_2) is given by $m = \frac{y_2 - y_1}{x_2 - x_1}$.

2. The line l has gradient $\frac{1}{3}$ and passes through $(0, 7)$. Find an equation for l in the form

a. $y = mx + c$

b. $px + qy + r = 0$

3. P The points $A(1, 2)$, $B(4, 1)$ and $C(12, k)$ are collinear. Work out the exact value of k.

Hint

A, B and C are **collinear** if they all lie on the same straight line.
This means that the gradients of the line segments AB, BC and AC are equal.

4. For the line with equation $4x - 5y + 12 = 0$, find:

a. the gradient

b. the coordinates of the y-intercept

c. the coordinates of the x-intercept.

Hint

The equation of a straight line can be written in the forms:

- $y = mx + c$, where m is the gradient and c is the y-intercept
- $ax + by + c = 0$, where a, b and c are integers.

5. E/P The gradient of the line joining the points $(2, 3a)$ and $(5a, -2)$ is -1.
Work out the value of a. (2 marks)

6. E/P The line l_1 with gradient $-\frac{1}{5}$ passes through $(0, 3)$. l_1 intersects the line l_2 with equation $3x - 4y + 7 = 0$ at point P. Find the exact coordinates of P. (4 marks)

7. E/P The points $A(-3p-3, 2p)$, $B(-5p+1, 0)$ and $C(0, 8p)$, where p is a constant, $p \neq 0$, are collinear.
Find:

a. the value of p (4 marks)

b. the gradient of the line through A, B and C. (2 marks)

8. E/P The line l_1 has gradient $\frac{2}{3}$ and passes through the point $(0, -4)$.

a. Find an equation of l_1 in the form $ax + by + c = 0$. (3 marks)

The line l_2 with equation $3x - ky + 25 = 0$ intersects l_1 at the point $(p, -2)$ where k and p are constants. Find:

b. the value of p (2 marks)

c. the value of k. (3 marks)

9. E/P The points A and B have coordinates $(-2, k+1)$ and $(3k-2, 6)$ where k is a constant. Given the gradient of AB is $-\frac{1}{2}$,

a. find the value of k. (3 marks)

The line through the points A and B passes through $(0, c)$. Find:

b. the value of c (3 marks)

c. the equation of the line in the form $ax + by + c = 0$. (3 marks)

5.2 Equations of straight lines

Practice Book, printed page 38–39.

1. Find the equation of the line with gradient m that passes through the point (x_1, y_1) when:

a. $m = -2$ and $(x_1, y_1) = (4, -5)$

b. $m = \frac{1}{4}$ and $(x_1, y_1) = (-2, 6)$

c. $m = -\frac{1}{8}$ and $(x_1, y_1) = (-3, -2)$

d. $m = -\frac{3}{5}$ and $(x_1, y_1) = (1, -3)$

Hint

The equation of the line with gradient m that passes through the point (x_1, y_1) can be found using $y - y_1 = m(x - x_1)$.

2. Find the equations of the lines that pass through these pairs of points:

a. $(4, -3)$ and $(-6, 9)$

b. $(-1, 3)$ and $(5, -2)$

c. $(\frac{1}{2}, -\frac{1}{3})$ and $(-\frac{3}{4}, \frac{2}{3})$

Hint

To find the equation of the line that passes through the points (x_1, y_1) and (x_2, y_2), first find the gradient m using $m = \frac{y_2 - y_1}{x_2 - x_1}$, then use $y - y_1 = m(x - x_1)$.

3. The line that passes through the points $(1, -4)$ and $(-3, 6)$ meets the x-axis at the point P. Work out the coordinates of P.

Hint

Find the equation of the line, then substitute in $y = 0$ and solve to find x.

4. The line that passes through the points $(-2, 5)$ and $(4, -3)$ meets the y-axis at the point Q. Work out the exact coordinates of Q.

5. E The line l passes through the points $A(3, -6)$ and $B(-2, -10)$. Find an equation for l, giving your answer in the form $y = mx + c$. (3 marks)

6. E The points $A(-6, 3)$ and $B(15, -4)$ lie on the line L. Find an equation for L in the form $ax + by + c = 0$, where a, b and c are integers. (3 marks)

7. E/P The line l_1 passes through the point $(6, -3)$ and has gradient $\frac{1}{3}$. l_1 meets the line l_2 with equation $x + 2y = 10$ at the point P. Calculate the coordinates of P. (4 marks)

8. E/P The line l_1 passes through the point $(5, -4)$ and has gradient $\frac{1}{4}$

a. Find an equation for l_1 in the form $ax + by + c = 0$, where a, b and c are integers. (3 marks)

The line l_2 passes through the origin O and has gradient -5. The lines l_1 and l_2 intersect at the point P.

b. Calculate the coordinates of P. (4 marks)

9. E/P The line l_1 passes through the points $A(-2, 3)$ and $B(4, -1)$.

a. Find an equation for l_1 in the form $ax + by + c = 0$, where a, b and c are integers. (4 marks)

The line l_2 with equation $3x + by - 1 = 0$ intersects l_1 at the point $P(k, k)$. Find:

b. the value of k (2 marks)

c. the value of b. (2 marks)

5.3 Parallel and perpendicular lines

Practice Book, printed page 39–40.

1. Work out whether these pairs of lines are parallel.

a. $2y = 3x - 5$
$6x - 4y + 11 = 0$

b. $3x - 4y + 9 = 0$
$9x + 12y - 10 = 0$

c. $5x + 2y - 15 = 0$
$10x + 4y + 9 = 0$

Hint

Parallel lines have the same gradient.

2. Work out whether each of these pairs of lines are perpendicular.

a. $3y = 2x + 7$
$4x + 6y + 1 = 0$

b. $5x - 3y + 2 = 0$
$5y = 3x + 6$

c. $4x - y - 5 = 0$
$2x + 8y - 15 = 0$

Hint

If two lines are perpendicular, the product of their gradients is -1.
If a line l has gradient m, the gradient of a line perpendicular to l is $-\frac{1}{m}$.

3. Find an equation of the line that passes through the point $(-3, 5)$ and is parallel to the line $2x - 3y + 7 = 0$.

4. Find an equation of the line that passes through the origin and is parallel to the line joining the points $(1, -6)$ and $(-2, 9)$.

5. E/P The line l_1 has equation $4y - 8 = 3x$. The point P with x-coordinate 4 lies on l_1.
The line l_2 is perpendicular to l_1 and passes through the point P. Find an equation of l_2, giving your answer in the form $ax + by + c = 0$, where a, b and c are integers. (4 marks)

6. E/P The line L has equation $2y = 1 - 3x$.

a. Show that the point $P(3, -4)$ lies on L. (1 mark)

b. Find an equation of the line perpendicular to L, which passes through P.
Give your answer in the form $ax + by + c = 0$, where a, b and c are integers. (3 marks)

7. E/P The line l_1 has equation $2x - 5y + 2 = 0$.

a. Find the gradient of l_1. (2 marks)

The line l_2 is perpendicular to l_1 and passes through the point $(3, 2)$.

b. Find the equation of l_2 in the form $ax + by + c = 0$, where a, b and c are integers. (3 marks)

8. E/P The line L_1 has equation $5y - 2x - k = 0$, where k is a constant.
Given that the point $A(1, 3)$ lies on L_1, find:

a. the value of k (1 mark)

b. the gradient of L_1. (2 marks)

The line L_2 passes through A and is perpendicular to L_1.

c. Find an equation of L_2, giving your answer in the form $ax + by + c = 0$, where a, b and c are integers. (3 marks)

The line L_2 crosses the x-axis at the point P.

d. Find the coordinates of P. (2 marks)

9. E/P The line l_1 passes through the points $P(-2, 3)$ and $Q(10, 9)$.

a. Find an equation for l_1 in the form $y = mx + c$, where m and c are constants. (3 marks)

The line l_2 passes through the point $R(12, 0)$ and is perpendicular to l_1.
The lines l_1 and l_2 intersect at the point S.

b. Calculate the coordinates of S. (5 marks)

5.4 Length and area

Practice Book, printed page 40–41.

1. Find the exact distance between these pairs of points:

a. $(1, -2)$, $(-5, 6)$

b. $(-8, 4)$, $(16, -3)$

c. $(6, -5)$, $(-10, -1)$

Hint

You can find the distance d between the points (x_1, y_1) and (x_2, y_2) by using the formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
If the number inside the square root is not a perfect square, write the surd in its simplest form.
← Section 1.5

2. Consider the points $A(-2, 5)$, $B(3, 1)$ and $C(8, -3)$. Determine whether the line segment joining the points A and B is congruent to the line segment joining the points B and C.

Hint

Line segments are congruent if they are the same length.

3. The vertices of a triangle are $P(-3, 2)$, $Q(2, 5)$ and $R(4, 2)$.
Find the area of triangle PQR.

Hint

Sketch the triangle and label its vertices.
Use area $= \frac{1}{2} \times$ base $\times$ perpendicular height.

4. P The distance between the points $(-2, 11)$ and $(x, 8)$ is $\sqrt{58}$.
Find the two possible values of x.

Hint

Use the formula for the distance between two points to form a quadratic equation, then solve this equation.
← Section 2.1

5. E The line l passes through the points $A(2, -6)$ and $B(-3, 14)$. Find:

a. an equation for l (3 marks)

b. the exact length of AB. (2 marks)

6. E/P The line l_1 passes through the point $A(4, 6)$ and has gradient $\frac{1}{2}$

a. Find an equation of l_1, giving your answer in the form $y = mx + c$. (3 marks)

The point B has coordinates $(-2, 3)$.

b. Show that B lies on l_1. (1 mark)

c. Find the length of AB, giving your answer in the form $k\sqrt{5}$ where k is a constant to be found. (2 marks)

The point C lies on l_1 and has x-coordinate equal to p. The length of AC is 6 units.

d. Show that p satisfies $5p^2 - 40p - 64 = 0$. (3 marks)

7. E The point $A(-6, 1)$ and the point $B(9, -4)$ lie on the line L.

a. Find an equation for L in the form $ax + by + c = 0$, where a, b and c are integers. (3 marks)

b. Find the distance AB, giving your answer in the form $k\sqrt{10}$, where k is an integer to be found. (2 marks)

8. E/P

a. Find an equation of the straight line passing through the points $(-3, 4)$ and $(5, -2)$. (3 marks)

Give your answer in the form $ax + by + c = 0$, where a, b and c are integers.

The line crosses the x-axis at point A and the y-axis at point B.

b. Find the area of triangle AOB, where O is the origin. (2 marks)

9. E/P The line l_1 passes through the points $P(-2, 4)$ and $Q(10, -2)$.

a. Find an equation for l_1 in the form $y = mx + c$, where m and c are constants. (3 marks)

The line l_2 passes through the point $R(0, -7)$ and is perpendicular to l_1.

The lines l_1 and l_2 intersect at the point S.

b. Calculate the coordinates of S. (5 marks)

c. Show that the length of RS is $4\sqrt{5}$. (2 marks)

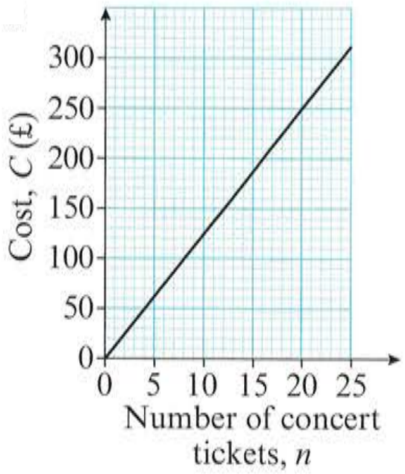
d. Hence, or otherwise, find the exact area of triangle PQR. (3 marks)

5.5 Modelling with straight lines

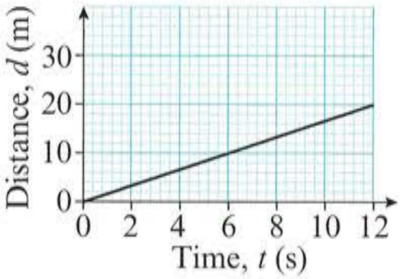
Practice Book, printed page 42–44.

1. For each graph:
i calculate the gradient, k, of the line
ii write down the units for the gradient
iii write a direct proportion equation connecting the two variables.

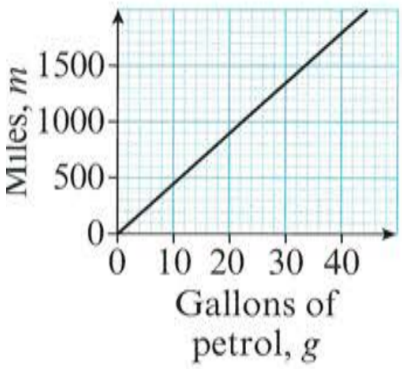
a



b



c



Hint

If two quantities, y and x, are in direct proportion they increase at the same rate.
The graph of y against x is a straight line which passes through the origin.
If y is directly proportional to x, then $y \propto x$ and $y = kx$ where k is a constant.
The units for the gradient are $\dfrac{\text{units of } y\text{-axis}}{\text{units of } x\text{-axis}}$.

2. Draw a graph to determine whether a linear model would be appropriate for each set of data.

a

p	q
0	0
2	4
4	9
6	15
10	20
20	42

b

l	v
0	25
5	30
10	45
20	80
30	160
40	280

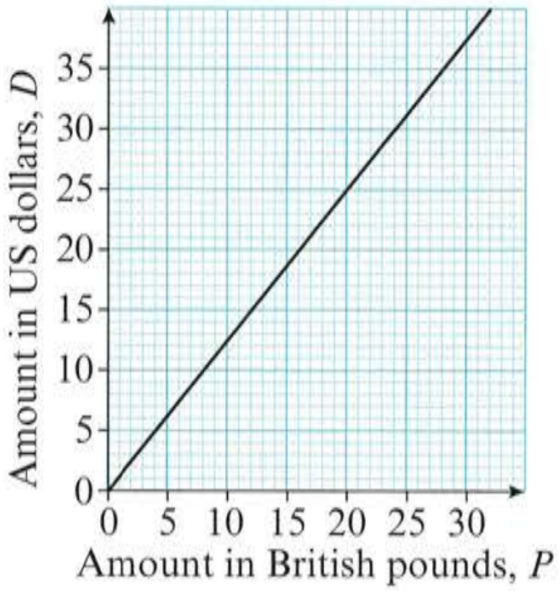
c

a	b
2.5	52
3.2	39
3.7	35
3.9	28
4.2	22
4.5	18

Hint

You can use a linear model for the relationship between two variables y and x if the points lie approximately on a straight line.

3. The graph shows the conversion between British pounds, P, and US dollars, D.



a. Calculate the gradient, k, of the line.

b. Write an expression for D in terms of P.

c. Explain what the value of k represents in this situation.

Hint

Use any two points on the line to calculate the gradient. As the line passes through the origin, the general form of the equation is $y = kx$.

4. Sand pours out from a 25 kg bag of sand through a hole at the bottom of the bag. The weight, w, of the bag of sand after t seconds is recorded in this table.

Weight, w kg	24.8	23.4	21.7	19.2	12.9	10.8
Time, t seconds	0	10	20	40	80	100

a. Plot the points on a graph to determine whether a linear model is appropriate.

b. Work out an equation for the line of best fit in the form $w = at + b$ where a and b are constants.

c. Interpret the meaning of the constants a and b.

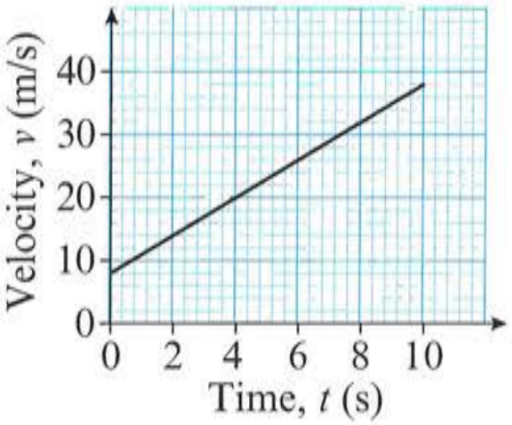
d. Use your model to estimate the time taken for the bag of sand to empty.

e. Suggest a reason your estimate may not be accurate.

Hint

You can choose two points from the table to work out the gradient for your linear model.
Make sure you give your answers in the context of the question.
Identify any assumptions you have made when using your linear model which may not be realistic.

5. E/P A car travels along a straight road. The graph shows the velocity, v m s^{-1}, of the car in the first 10 seconds of its motion.



a. Find a linear equation of the form $v = a + bt$ linking v and t, where t is the time in seconds. (3 marks)

b. Interpret the values of a and b. (2 marks)

c. Use the model to predict the velocity of the car after 15 seconds. (1 mark)

d. Write down one reason why this might not be a realistic model. (1 mark)

6. E/P The table shows the times achieved by gold medallists in the women's 100 m sprint in the Olympic Games from 1992 to 2016.

Year, Y	1992	1996	2004	2008	2012	2016
Time, T **seconds**	10.82	10.94	10.93	10.78	10.75	10.71

a. Draw a scatter graph to show these results. (3 marks)

b. Use the points $(1996, 10.94)$ and $(2016, 10.71)$ to formulate a linear model for this data. (3 marks)

c. Use your model to predict the gold-medal winning time in the women's 100 m in the 2020 Olympic Games. (1 mark)

d. Comment on the validity of this model for predicting winning times. (1 mark)

7. E/P The charge for a taxi journey is made up of a fixed charge plus a further cost per mile travelled. A taxi company charges £20.40 for a 5-mile journey and £56.40 for a 15-mile journey.

a. Write an equation linking the charge, £C, with the number of miles, m, in the form $C = am + b$ (3 marks)

b. Interpret the values of a and b in the context of the question. (2 marks)

c. Work out the taxi fare for a 12-mile journey. (1 mark)

8. E/P A man buys a new motorbike for £5475. After 1 year the motorbike is worth £4380.

a. Find a linear model to link the value of the motorbike, £V, with the age of the motorbike, t years. (3 marks)

b. Interpret the gradient of your model. (1 mark)

c. Use your model to predict the value of the motorbike after 2.5 years. (1 mark)

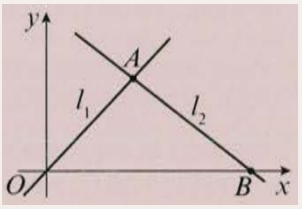
d. Comment on the suitability of the model. (1 mark)

Problem solving: Set A

Practice Book, printed page 44.

Bronze

The line l_1 passes through the origin O and $A(3, 6)$ as shown in the diagram.
The length of OA is $a\sqrt{5}$.



a. Find the value of a. (2 marks)

The line l_2 is perpendicular to l_1, passes through A and crosses the x-axis at the point B. Find:

b. the gradient of l_2 (2 marks)

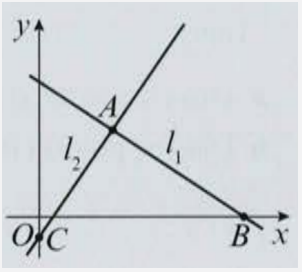
c. an equation for l_2, giving your answer in the form $ax + by + c = 0$ (3 marks)

d. the coordinates of B (1 mark)

e. the area of $\triangle ABC$. (3 marks)

Silver

The points $A(2, 3)$ and $B(8, 0)$ lie on the line l_1, as shown in the diagram.
The line l_2 is perpendicular to l_1, passes through A and crosses the y-axis at the point C.

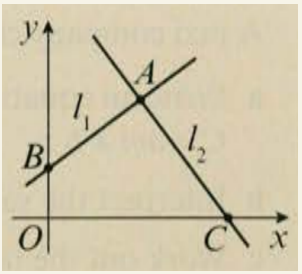


a. Find the coordinates of C. (5 marks)

b. Hence find the area of $\triangle ABC$. (2 marks)

Gold

The points $A(2, 9)$ and $B(0, 5)$ lie on the line l_1, as shown in the diagram.
The line l_2 is perpendicular to l_1, passes through A and crosses the x-axis at the point C.
Find the area of $\triangle ABC$. (8 marks)



Problem solving: Set B

Practice Book, printed page 45.

Bronze

The point P has coordinates $(7, 1)$. The distance from P to the point $R(10, k)$ is 4.

a. Show that $k^2 - 2k - 6 = 0$. (3 marks)

b. Given that k is positive, find the exact value of k. (3 marks)

Silver

The point P has coordinates $(3, -4)$. The distance from P to the point $R(k, 2)$ is 10.
Find the two possible values of k. (6 marks)

Gold

The point P has a positive x-coordinate and lies on the line with equation $y = x$.
The point Q has coordinates $(0, 4)$. Given that the distance PQ is 6, find the exact coordinates of P. (8 marks)

Selected exam question bank questions

Question 21

The line l_1 has the equation $2y - 9 = 5x$. The point A with x-coordinate 3 lies on l_1.

The line l_2 is perpendicular to l_1 and passes through the point A.

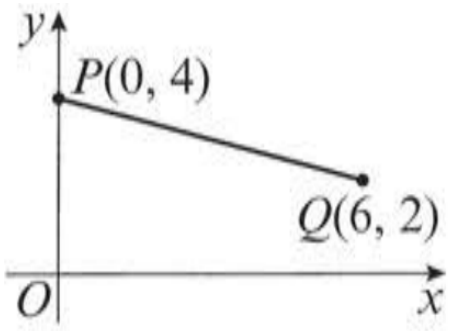
a. Find an equation for l_2 in the form $ax + by + c = 0$. (4 marks)

The lines l_1 and l_2 cross the x-axis at the points B and C respectively.

b. Calculate the area of the triangle ABC. (4 marks)

Question 77

The diagram shows the line segment joining the points $P(0, 4)$ and $Q(6, 2)$.



a. Find the equation of the line perpendicular to PQ that passes through the origin, O. (2 marks)

b. Find the coordinates of the point where this perpendicular meets PQ. (4 marks)

c. Show that the perpendicular distance of PQ from the origin is $\frac{6\sqrt{10}}{5}$. (2 marks)

d. Find the length of PQ giving your answer in the form $a\sqrt{10}$. (2 marks)

e. Find the area of triangle OPQ. (2 marks)

Question 81

The line l_1 with equation $y = 2x + 5$ intersects the line l_2 with equation $4x + 3y - 35 = 0$ at the point P.

a. Find the coordinates of P. (5 marks)

The lines l_1 and l_2 cross the line $y = 1$ at the points Q and R respectively.

b. Find the area of triangle PQR. (4 marks)

The distance of the point $A(10, k)$ from P is less than 10.

c. Find the range of possible values for k. (4 marks)

Question 86

The line l_1 passes through the points $A(4, -1)$ and $B(-2, 7)$.

a. Find an equation of l_1, giving your answer in the form $ax + by + c = 0$. (4 marks)

b. Find an equation of the line l_2 which passes through B and is perpendicular to l_1. (3 marks)

The point C has coordinates $(k, 1)$ and angle ABC is a right angle.

c. Find the value of k. (2 marks)

d. Find the equation of the circle that passes through the points A, B and C. (5 marks)

Question 89

The points P, Q and R have coordinates $(6, 2)$, $(k, 8)$ and $(9, 3)$ respectively. Given that the distance between points P and Q is twice the distance between points P and R,

a. calculate the possible values of k. (7 marks)

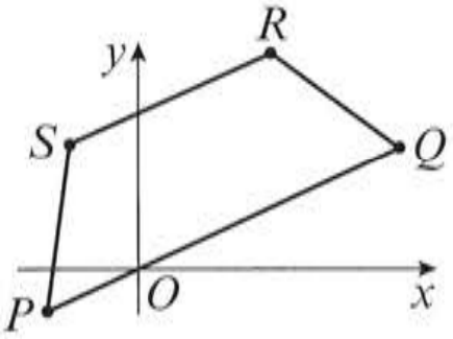
Given that the line passing through P and Q has equation $3x - y - 16 = 0$,

b. find the coordinates of the midpoint of PQ. (4 marks)

Question 93

The diagram shows a trapezium $PQRS$.

The coordinates of the vertices are $P(-10, -6)$, $Q(15, 9)$, $R(10, 16)$ and $S(-5, 7)$.



a. Verify that the lines PQ and SR are parallel. (3 marks)

b. Prove that the trapezium is not isosceles. (3 marks)

c. The diagonals of the trapezium meet at M. Find the coordinates of M. (4 marks)

d. Show that neither of the diagonals of the trapezium bisects the other diagonal. (3 marks)

Answers

5.1 $y = mx + c$

1. **a** $\frac{3}{4}$ **b** $-\frac{3}{2}$ **c** $\frac{2}{5}$
2. **a** $y = \frac{1}{3}x + 7$ **b** $x - 3y + 21 = 0$
3. $-\frac{5}{3}$
4. **a** $\frac{4}{5}$ **b** $(0, \frac{12}{5})$ **c** $(-3, 0)$
5. 2
6. $(\frac{25}{19}, \frac{52}{19})$
7. **a** 5 **b** $\frac{5}{3}$
8. **a** $2x - 3y - 12 = 0$ **b** 3 **c** -17
9. **a** -10 **b** -10 **c** $x + 2y + 20 = 0$

5.2 Equations of straight lines

1. **a** $y = -2x + 3$ **b** $4y = x + 26$ **c** $8y = -x - 19$ **d** $5y = -3x - 12$
2. **a** $5y = -6x + 9$ **b** $6y = -5x + 13$ **c** $15y = -12x + 1$
3. $(-\frac{3}{5}, 0)$
4. $(0, \frac{7}{3})$
5. $y = \frac{4}{5}x - \frac{42}{5}$
6. $x + 3y - 3 = 0$
7. $(12, -1)$
8. **a** $x - 4y - 21 = 0$ **b** $(1, -5)$
9. **a** $2x + 3y - 5 = 0$ **b** 1 **c** -2

5.3 Parallel and perpendicular lines

1. **a** Parallel **b** Not parallel **c** Parallel
2. **a** Not perpendicular **b** Not perpendicular **c** Perpendicular
3. $3y = 2x + 21$
4. $y + 5x = 0$
5. $4x + 3y - 31 = 0$
6. **a** $2 \times (-4) = 1 - 3 \times 3 = -8$ **b** $2x - 3y - 18 = 0$
7. **a** $\frac{2}{5}$ **b** $5x + 2y - 19 = 0$
8. **a** 13 **b** $\frac{2}{5}$ **c** $5x + 2y - 11 = 0$ **d** $(\frac{11}{5}, 0)$
9. **a** $y = \frac{1}{2}x + 4$ **b** $(8, 8)$

5.4 Length and area

1. **a** 10 **b** 25 **c** $4\sqrt{17}$

2. Congruent. $AB = BC = \sqrt{41}$

3. 10.5

4. 5 and -9

5. **a** $y = 2 - 4x$ **b** $5\sqrt{17}$

6.
- **a.** $y = 0.5x + 4$
- **b.** $x = -2$, $y = 0.5 \times (-2) + 4 = 3$
- **c.** $AB = 3\sqrt{5}$
- **d.** $(p-4)^2 + (0.5p-2)^2 = 36$
  $\Rightarrow p^2 - 8p + 16 + \frac{1}{4}p^2 - 2p + 4 = 36$
  $\Rightarrow 5p^2 - 40p - 64 = 0$

7. **a** $x + 3y + 3 = 0$ **b** $5\sqrt{10}$

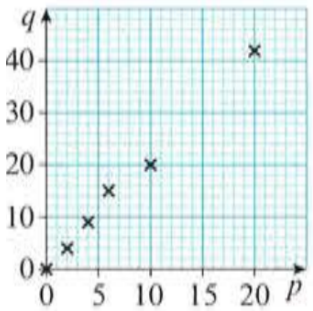
8. **a** $3x + 4y - 7 = 0$ **b** $\frac{49}{24}$

9. **a** $y = 3 - 0.5x$ **b** $(4, 1)$
c $\sqrt{(4-0)^2 + (1+7)^2} = 4\sqrt{5}$ **d** 60

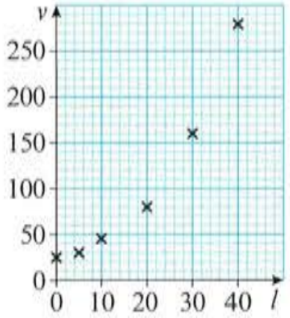
5.5 Modelling with straight lines

1.
- **a.** **i** $k = \frac{25}{2}$ **ii** cost per concert ticket **iii** $C = \frac{25}{2}n$
- **b.** **i** $k = \frac{5}{3}$ **ii** m s^{-1} **iii** $d = \frac{5}{3}t$ or $3d = 5t$
- **c.** **i** $k = 45$ **ii** miles per gallon **iii** $m = 45g$

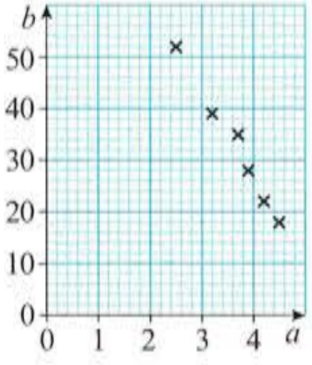
2. **a** linear



b non-linear

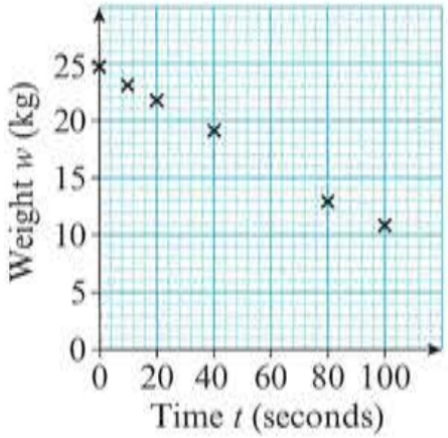


c linear



3. **a** 1.25 **b** $D = 1.25P$
c The number of US dollars per British pound.

4. **a.**



A linear model is appropriate.

b. $w = -0.14t + 24.8$
Answers may vary slightly depending on which points of the graph were used.

c. $a = -0.14$ is the number of kg the weight goes down by per second. $b = 24.8$ is the initial weight of the bag of sand.

d. Approx 177 seconds
Answers may vary slightly depending on which points of the graph were used.

e. The estimate assumes that rate of flow of sand is constant. As the bag empties, the reduced weight of sand in the bag may cause the sand to flow at a slower rate.

5.

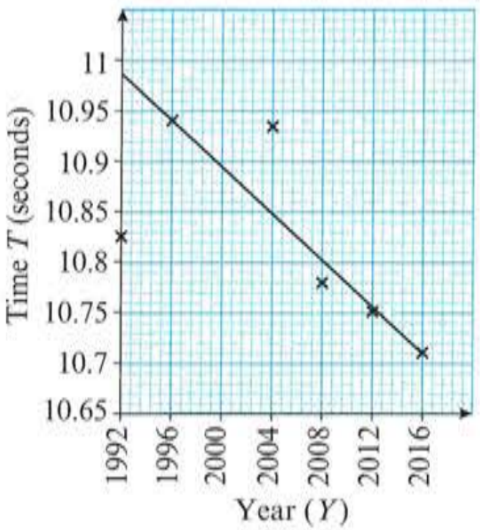
a. $v = 8 + 3t$

b. $a = 8$ m s^{-1} is the velocity when $t = 0$; $b = 3$ is the rate of change of velocity (i.e. the acceleration).

c. 53 m s^{-1}

d. The model suggests the speed will continue to increase indefinitely at a constant rate, which is not possible.

6. **a.**



b. $T = 33.894 - 0.0115Y$

c. 10.66 seconds

d. E.g. the model is not valid as it would eventually predict times $\leqslant 0$ to complete the race.

7.

a. $C = 3.6m + 2.4$

b. a represents a cost of £3.60 per mile and b represents the £2.40 standing charge.

c. £45.60

8.

a. $V = 5475 - 1095t$

b. The rate of change of the value of the motorbike, i.e. a decrease of £1095 per year.

c. £2737.50

d. The model suggests that the value will be negative after some years ($t > 5$), which is not realistic.

Problem solving: Set A

Bronze **a** 3 **b** $-\frac{1}{2}$ **c** $x + 2y - 15 = 0$ **d** $(15, 0)$ **e** 45

Silver **a** $(0, -1)$ **b** 15

Gold 45

Problem solving: Set B

Bronze **a** $(10 - 7)^2 + (k - 1)^2 = 16 \Rightarrow 9 + k^2 - 2k + 1 = 16 \Rightarrow k^2 - 2k - 6 = 0$

b $k = 1 + \sqrt{7}$

Silver 11 and -5

Gold $(2 + \sqrt{14}, 2 + \sqrt{14})$

Selected exam question bank answers

Question 21 **a** $2x + 5y - 66 = 0$ **b** 208.8

Question 77

a. $y = 3x$

b. $(\frac{6}{5}, \frac{18}{5})$

c. $\sqrt{(\frac{6}{5})^2 + (\frac{18}{5})^2} = \sqrt{\frac{360}{25}} = \frac{6\sqrt{10}}{5}$

d. $2\sqrt{10}$

e. 12

Question 81 **a** $(2, 9)$ **b** 40 **c** $3 < k < 15$

Question 86 **a** $4x + 3y - 13 = 0$ **b** $3x - 4y + 34 = 0$

c -10 **d** $(x + 3)^2 + y^2 = 50$

Question 89 **a** 4 and 8 **b** $(7, 5)$

Question 93

a. Gradient of $PQ = \frac{15}{25} = \frac{3}{5}$; gradient of $SR = \frac{9}{15} = \frac{3}{5}$
So PQ and SR are parallel.

b. Length of $PS = \sqrt{194}$; length of $QR = \sqrt{74}$
So trapezium is not isosceles.

c. $(\frac{5}{2}, \frac{31}{4})$

d. Midpoint of PR is $(0, 5)$; midpoint of SQ is $(5, 8)$, so neither diagonal of the trapezium bisects the other.